



PROJECT



COURSE INFORMATION

- ❖ Course Code: CSE200
- ❖ Course Title: Computer Aided Engineering Drawing
- ❖ Topic: Design of a single studio apartment



SUBMITTED TO

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1. Introduction

The purpose of this project is to design a functional and efficient studio apartment within a limited space of 28 ft × 22 ft. The project focuses on integrating essential living areas such as sleeping, kitchen, dining, and washroom into a compact layout while maintaining comfort, usability, and aesthetic appeal.

The scope of this project includes both 2D floor planning and 3D visualization, ensuring that the design is practical and structurally feasible.

2. Project Objectives

- To design a compact yet functional studio apartment layout
- To ensure proper space utilization and efficient circulation
- To include all essential living spaces within limited dimensions
- To create both 2D and 3D representations of the design
- To develop skills in architectural planning and visualization

3. Project Description

The project involves designing a studio apartment that combines living and sleeping areas while incorporating separate functional spaces such as a kitchen, washroom, and dining area.

The design emphasizes simplicity, functionality, and efficient use of space. Natural lighting and ventilation are considered by placing windows strategically. The layout ensures smooth movement between spaces and maintains privacy where needed.

4. Design Specifications

- **Total Area:** 28 ft × 22 ft
- **Living & Sleeping Area:** 14 ft × 12 ft
- **Kitchen Area:** 8 ft × 6 ft
- **Washroom:** 5 ft × 4 ft
- **Dining Area:** 6 ft × 6 ft

Structural Details:

- Wall thickness: 5–6 inches (standard residential wall)
- Door height: 7 ft
- Window size: 4 ft × 3 ft (standard)
- Ceiling height: 10 ft

Layout Requirements:

- One main entrance
- Windows for natural light and ventilation
- Efficient placement of furniture and fixtures

5. 2D Floor Plan Description

The 2D floor plan presents a top view of the apartment layout with proper dimensions and labels.

- The **main entrance** opens into the living/sleeping area
- The **living and sleeping area (14 × 12 ft)** serves as the central space
- The **kitchen (8 × 6 ft)** is positioned adjacent to the main area with a counter layout
- The **washroom (5 × 4 ft)** is placed in a corner for privacy
- The **dining area (6 × 6 ft)** is located near the kitchen for convenience

All areas are clearly labeled, and dimensions are properly marked to ensure accuracy and readability.

6. 3D Model Description

The 3D model provides a realistic visualization of the apartment design.

- **Wall Height:** 10 ft
- **Roof Design:** Flat roof (modern style)
- **Interior Visualization:**
 - Furniture placement (bed, sofa, dining table)
 - Kitchen cabinets and counter
 - Washroom fixtures

The model helps in understanding spatial relationships, lighting, and overall aesthetics. It enhances the presentation and provides a real-life feel of the design.

7. Tools and Software Used

The following tools were used for designing and modeling:

- AutoCAD (for 2D floor plan)
- SketchUp / Blender (for 3D modeling)
- Microsoft Word (for report preparation)

These tools helped in creating accurate designs and professional presentations.

8. Challenges and Solutions

Challenges Faced:

- Limited space management
- Proper placement of all functional areas
- Maintaining ventilation and lighting

Solutions:

- Used compact and multi-functional furniture
- Optimized layout for better space utilization
- Added windows strategically for airflow and light

9. Conclusion

This project successfully demonstrates the design of a compact studio apartment with efficient space utilization. The integration of 2D and 3D designs helped in visualizing and refining the layout.

Through this project, important skills in planning, designing, and problem-solving were developed. The final design meets all functional and aesthetic requirements within limited space.